

Notes: Bloom, Sadun, and Van Reenen (QJE, 2012)
The Organization of Firms Across Countries

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1 Big picture

- **Question.** Does social capital, proxied by trust, affect productivity by changing how firms are organized internally?
- **Core mechanism.** The paper argues that trust makes headquarters more willing to delegate real authority to plant managers. Delegation allows CEOs to leverage their time over more plants and employees.
- **Main outcome.** The central organizational outcome is decentralization: how much autonomy plant managers have over investment, hiring, product, and sales or marketing decisions.
- **Data contribution.** The authors collect a new cross-country survey of almost 4,000 medium-sized manufacturing firms in the United States, Europe, and Asia. The survey measures internal organization in a comparable way across countries.
- **Main empirical fact.** Firms headquartered in high-trust regions are more decentralized. This holds after extensive controls for country, industry, region, skills, size, ownership, and survey noise.
- **Multinational evidence.** Within multinational firms, subsidiaries are more decentralized when the parent-country population has higher bilateral trust toward the subsidiary-country population.
- **Identification strategy.** The paper instruments bilateral trust using religious similarity between country pairs. The IV evidence supports a causal interpretation: trust increases decentralization.
- **Firm-size channel.** Higher trust is also associated with larger firms. The theory says this happens because delegation economizes on CEO time and allows productive firms to grow.
- **Aggregate implication.** Trust can raise aggregate productivity by enabling better reallocation: efficient firms can expand, take market share, and operate at larger scale.
- **Economic takeaway.** Social capital matters not only through contracts, finance, or trade. It also matters inside the firm, by shaping whether authority can be credibly delegated.

2 Incremental contribution of the paper

- **From trust and growth to a firm-level mechanism.** Prior work links trust to country-level growth and large organizations. This paper opens the mechanism by showing that trust changes the internal allocation of authority inside firms.
- **A new international organization data set.** The paper provides comparable firm-level data on decentralization across 12 countries. This is a major empirical contribution because organizational economics had much more theory than cross-country micro data.
- **A theory linking trust, delegation, and firm size.** The paper adapts the Garicano hierarchy model. Trust increases the expected payoff from letting plant managers solve problems, which increases decentralization and firm scale.

- **Within-country and within-multinational identification.** The paper goes beyond cross-country correlations by using regional variation in trust and multinational subsidiaries with different parent-country trust environments.
- **Bilateral trust as an organizational determinant.** The paper shows that trust between two countries affects not only trade and investment flows but also the internal authority structure of multinational affiliates.
- **Religious similarity instrument.** The use of religious similarity as an instrument for bilateral trust gives a historical-cultural source of variation that is plausibly less related to current firm-level decentralization shocks.
- **Link to reallocation and productivity.** The paper connects organization to aggregate productivity: decentralization helps productive firms become large, allowing resources to move toward more efficient producers.
- **Controls for management quality.** Because better managed firms may both decentralize and locate in high-trust areas, the paper controls for Bloom and Van Reenen management scores. The trust effect remains.
- **Broader comparative advantage implication.** High-trust areas may attract or support industries that require more decentralization, suggesting a cultural-organizational channel of comparative advantage.

3 Data and measurement

3.1 Organization survey

- The survey covers medium-sized manufacturing firms with 100 to 5,000 workers.
- The sample spans 12 countries: China, France, Germany, Greece, India, Italy, Japan, Poland, Portugal, Sweden, the United Kingdom, and the United States.
- The baseline regressions use 3,655 firm observations.
- Interviews were conducted by telephone with plant managers, usually senior enough to understand plant operations but close enough to daily decisions to report actual practices.
- The median firm is medium-sized; the authors intentionally avoid very small firms with little public data and very large firms whose internal structure may vary too much across plants.

Interpretation: the survey is designed to capture actual authority inside comparable manufacturing firms rather than broad perceptions about national culture.

3.2 Decentralization index

The paper asks plant managers how much authority they have in four areas:

- **Capital investment.** How much investment can the plant manager undertake without prior authorization from headquarters?
- **Hiring.** Where is the decision made to hire a full-time permanent shop-floor employee?

- **Product introduction.** Where are decisions made about introducing new products?
- **Sales and marketing.** Where are sales and marketing decisions made?

The qualitative questions are scored from 1 to 5, where 1 means decisions are made at corporate headquarters and 5 means the plant manager has real authority. The investment question is continuous and converted to dollars using purchasing power parity. Each component is standardized as a z-score, and the main decentralization measure is the unweighted average of the four z-scores.

Interpretation: decentralization is measured as the real authority of the plant manager relative to corporate headquarters, not as the number of hierarchical layers.

3.3 Survey quality and measurement controls

- The survey is **double blind**: managers do not know they are being scored, and interviewers do not know the firm’s performance.
- Questions are open-ended, so interviewers score actual practices rather than simple yes/no claims.
- Interviewers were trained together and worked from one location to improve cross-country comparability.
- The regressions include interviewer fixed effects, interview duration, reliability scores, day-of-week controls, manager seniority, manager tenure, and other survey-noise controls.
- Repeat interviews with 72 firms produce a correlation of 0.513 across independent interviews, suggesting meaningful firm-level signal but also measurement error.

Interpretation: the authors devote substantial effort to making an unusually difficult organizational variable comparable across countries.

3.4 Trust measures

The main trust measure comes from the World Values Survey question:

“Generally speaking, would you say that most people can be trusted, or that you can’t be too careful in dealing with people?”

The trust variable is the percentage of respondents in the relevant region or country answering “most people can be trusted.” For domestic firms, the paper uses trust in the headquarters region. For foreign multinationals, it uses the trust level in the parent firm’s country of origin.

For multinational subsidiaries, the paper also uses **bilateral trust**: how much people from the parent country say they trust people from the subsidiary country. This is especially useful because it varies by country pair.

Interpretation: the key theoretical object is the trust of headquarters toward local managers. The headquarters-region and bilateral-trust measures are attempts to proxy that object.

3.5 Main controls

The regressions include several layers of controls:

- plant skills, measured by the share of employees with a college degree;
- firm employment and plant employment as a share of firm employment;
- foreign and domestic multinational indicators;
- public listing status;
- whether the CEO is on the same site as the plant;
- fraction of managers native to the plant country;
- GDP per capita and population in relevant locations;
- country dummies, region controls, industry dummies, and survey-noise controls.

Interpretation: the empirical design tries to separate trust from size, human capital, development, ownership form, measurement noise, and broad country-level factors.

4 Theory and empirical predictions

4.1 Hierarchy model with trust

The theory builds on Garicano’s model of hierarchies and knowledge. Firms face production problems. Plant managers can solve common problems if trained; rare problems are passed up to the CEO. Delegation saves CEO time but is costly because managers must know what to do and because the CEO must trust them to act correctly.

Let z_M denote the mass of problems the plant manager is trained to solve. Let $F(z_M)$ be the share of problems covered by this training. Let $\lambda \in (0, 1]$ denote trust: the probability that the manager solves delegated problems correctly.

Production per plant is:

$$F(z_M)\lambda + (1 - F(z_M)).$$

The first term is the share of problems handled by the plant manager times the probability that they are handled correctly. The second term is the share of problems passed to the CEO, who can solve them correctly.

Interpretation: when trust is low, delegation is risky because the manager may misuse authority or make decisions inconsistent with headquarters’ objectives.

4.2 CEO time constraint

The CEO spends communication time h on problems that plant managers do not solve. With n plant managers, the CEO time constraint is:

$$(1 - F(z_M))nh = 1.$$

The firm chooses decentralization z_M and firm scale n . More delegation means plant managers solve more problems, so the CEO can supervise more plant managers.

Interpretation: the scarce resource is not capital alone, but senior management time. Trust expands the effective span of control of the CEO.

4.3 Proposition 1: trust increases decentralization

The first-order condition implies that higher trust raises the optimal degree of decentralization:

$$\frac{\partial z_M^*}{\partial \lambda} > 0.$$

The intuition is simple. When plant managers are more likely to take correct actions, the marginal benefit of delegating additional problems to them rises.

Interpretation: trust does not merely make people feel better. It changes the optimal allocation of decision rights.

4.4 Proposition 2: trust increases firm size

Firm size is:

$$n^* = \frac{1}{(1 - F(z_M^*))h}.$$

Because trust increases z_M^* , it reduces the number of problems the CEO must handle directly. Hence:

$$\frac{\partial n^*}{\partial \lambda} > 0.$$

Interpretation: high trust allows talented CEOs to leverage their managerial ability over more plants and workers. This produces larger firms and potentially greater aggregate productivity through reallocation.

4.5 Skill, wages, and selection

The model also implies that more skilled plant managers should receive more authority because it is less costly to train them to handle problems. In general equilibrium, higher trust may raise wages or housing costs, which partially offsets the firm-size effect. But selection reinforces the mechanism: high-trust regions allow more productive firms to expand and less productive firms to exit.

Interpretation: the paper's firm-size result is not an isolated prediction. It follows naturally from delegation, span of control, and reallocation.

5 Empirical specifications

5.1 Trust and decentralization regression

The baseline decentralization regression is conceptually:

$$Decentralization_i = \alpha + \beta Trust_{CHQ(i)} + X_i' \Gamma + \delta_c + \delta_j + \varepsilon_i,$$

where $Trust_{CHQ(i)}$ is trust in the headquarters region or country, X_i contains firm, plant, and survey controls, δ_c are country or location controls, and δ_j are industry effects.

The coefficient of interest is β . The model predicts $\beta > 0$.

Interpretation: the regression asks whether firms whose headquarters are embedded in higher-trust environments give more authority to plant managers.

5.2 CEO on-site versus off-site

The paper splits the sample according to whether the CEO is located at the same site as the plant manager. The model predicts that trust should matter more when the CEO is off-site, because direct monitoring is harder and communication costs are higher.

Interpretation: this is a mechanism test. If trust mainly helps when monitoring is difficult, the trust coefficient should be stronger when the CEO is physically separated from the plant manager.

5.3 Multinational and bilateral-trust specification

For multinational subsidiaries, the paper estimates regressions using trust from the parent-country perspective:

$$Decentralization_{ipc} = \alpha + \beta BilateralTrust_{origin,location} + X_{ipc}' \Gamma + \delta_{origin} + \delta_{location} + \delta_j + \varepsilon_{ipc}.$$

The strongest specifications include both country-of-origin and country-of-location fixed effects. Identification then comes from pairwise variation in trust between countries.

Interpretation: this design compares, for example, how much authority a Swedish-owned affiliate and a French-owned affiliate give to plant managers in the same host country, and how that relates to bilateral trust.

5.4 Instrumental variables

The paper instruments bilateral trust with religious similarity between country pairs. The first stage is strong: religious similarity predicts bilateral trust. The second stage asks whether the predicted component of trust raises decentralization.

Interpretation: religious similarity is intended to capture deep historical-cultural proximity that affects trust but is less likely to be caused by current decentralization choices.

5.5 Firm size regression

The firm-size regressions test Proposition 2:

$$\log(\textit{Employment})_r = \alpha + \beta \textit{Trust}_r + \textit{controls} + \varepsilon_r.$$

For multinationals, the paper uses bilateral-trust variation across country-of-origin and country-of-location pairs and considers mean employment, total employment, and number of subsidiaries.

Interpretation: if trust facilitates decentralization, it should also support larger firms or larger multinational footprints.

6 Identification and empirical design

6.1 What the design can identify

The paper is strongest as evidence that trust is systematically related to internal organization and likely has a causal effect on delegation. The identification strategy combines several sources of variation:

- regional variation in trust within countries;
- country fixed effects to remove broad institutional differences;
- plant-location fixed effects in multinational samples;
- country-of-origin fixed effects for parent-country characteristics;
- bilateral trust, which varies by country pair;
- religious similarity as an instrument for bilateral trust.

Interpretation: the empirical design moves from broad correlation to more credible causal evidence by comparing firms operating in similar environments but carrying different trust cultures from their headquarters.

6.2 Main threats

- **Omitted institutions.** High trust may proxy for better legal systems, lower regulation, or higher development. The paper addresses this with rule-of-law controls, GDP controls, population controls, and country dummies.
- **Management quality.** Better managed firms may be both more decentralized and more common in high-trust areas. The paper controls for the Bloom-Van Reenen management score.
- **Human capital.** Skilled workers may allow decentralization and may also be concentrated in high-trust areas. The paper controls for plant skills and regional development.
- **Ownership and family firms.** Low-trust environments may produce more family firms, which tend to centralize. The robustness checks include family management.
- **Measurement error.** Decentralization and trust are difficult to measure. The paper uses careful survey design, repeat interviews, and extensive noise controls.

- **Broader cultural proximity.** Religious similarity may affect decentralization through channels other than trust. The paper reports robustness checks with gravity and cultural controls in the online appendix.

Interpretation: no single test is definitive, but the consistency across region, multinational, IV, and robustness specifications makes the trust-decentralization interpretation persuasive.

6.3 Interpreting the coefficient sizes

In the simplest baseline, a one-standard-deviation increase in trust is associated with about 0.15 standard deviations higher decentralization. With full covariates, the effect is about 0.07 standard deviations. Moving from a very low-trust region to a very high-trust region is associated with a much larger decentralization difference.

For firm size, the estimates imply economically large effects. A one-standard-deviation increase in regional trust is associated with roughly a 30 percent increase in firm size in the specification with country dummies.

Interpretation: the magnitudes are large enough to matter for the size distribution of firms and, through reallocation, for aggregate productivity.

7 Tables and figures

7.1 Figure I: examples of firm organizational structures

Read:

- Figure I explains how the paper defines decentralization across different organizational forms.
- Example A is a simple domestic firm with one headquarters and one plant. Decentralization is the authority of the plant manager relative to headquarters.
- Example B is a multi-site domestic firm. The authors usually observe one plant and assume it is representative of the firm’s broader decentralization.
- Example C shows a single-site firm where headquarters and plant are co-located. The authors keep these observations only if the plant manager is not the CEO.
- Example D shows a multinational. Decentralization is measured from the global headquarters to the local plant manager.

Economic message: The figure clarifies the core object of measurement: real authority from corporate headquarters to plant managers. This is essential because the paper’s trust mechanism concerns delegation across hierarchical levels.

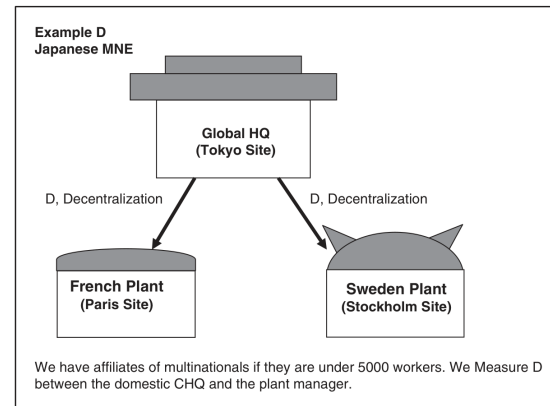
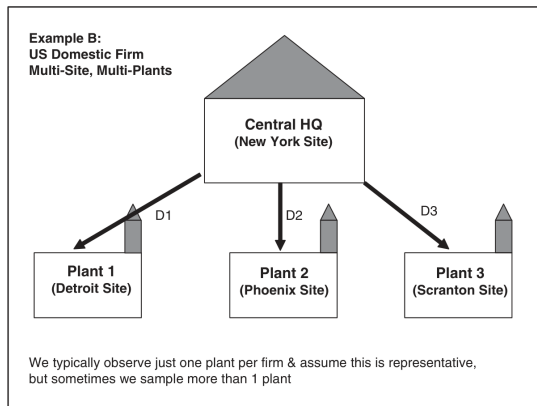
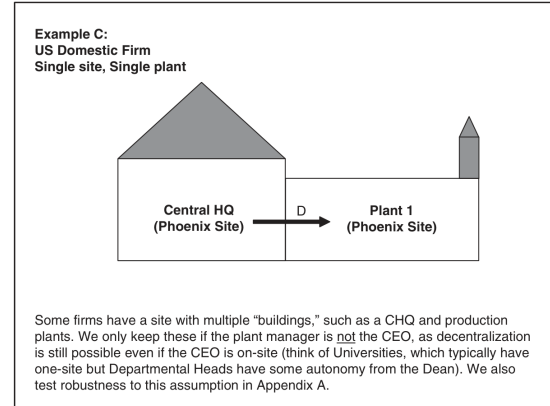
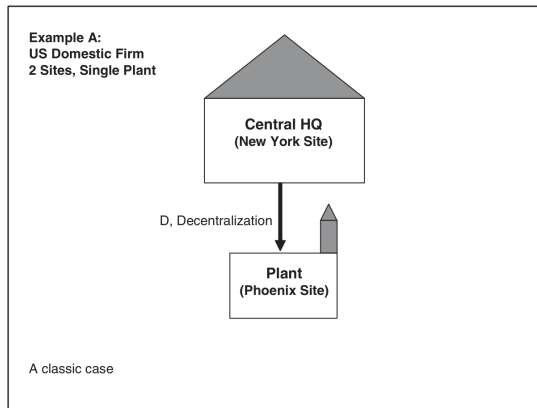


FIGURE I
Examples of Firm Organizational Structures

FIGURE I
Continued

(a) Figure I, part 1: domestic single-plant and multi-plant cases

(b) Figure I, part 2: same-site and multinational cases

7.2 Figures II and III: trust and decentralization by country

Read:

- Figure II shows large cross-country differences in trust. Portugal, France, and Greece have low trust; Sweden, China, and the United States appear high in the plotted distribution.
- The figure also shows meaningful regional variation within countries, which is important for the within-country empirical design.
- Figure III shows decentralization by country. Greece, Japan, India, and China are relatively centralized; Sweden, the United States, the United Kingdom, and Germany are relatively decentralized.
- The ordering is not identical to trust, but the broad positive association is visible: high-trust countries tend to have more decentralized firms.

Economic message: The raw data already point to the main hypothesis. Trust and decentralization differ substantially across countries, and higher trust tends to coincide with greater plant-manager autonomy.

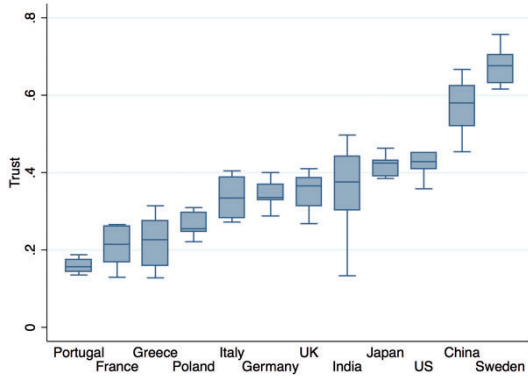


FIGURE II
Trust by Country (across Regions)

The graph shows levels of regional trust by country at the 25th percentile (bottom line of the box), median (middle line of the box), and 75th percentile (top line of the box). Upper and lower adjacent values are also shown. Regional trust measures are computed by pooling four successive waves of the World Values Survey (1981–1984, 1989–1993, 1994–1999, and 1999–2004), with the exception of Greece, for which the trust measure is built from the European Social Survey (all waves between 2000 and 2005). Trust is the percentage of people in the region answering “Most people can be trusted” to the question “Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?” with the alternative being “Can’t be too careful.” The number of responses used to generate the trust measure for each country is as follows: Portugal = 2,124; France = 2,499; Greece = 4,972; Poland = 2,768; Italy = 3,877; Germany = 7,870; United Kingdom = 3,434; India = 6,032; Japan = 4,254; United States = 4,410; China = 2,047; Sweden = 1,918.

(a) Figure II: trust by country, across regions

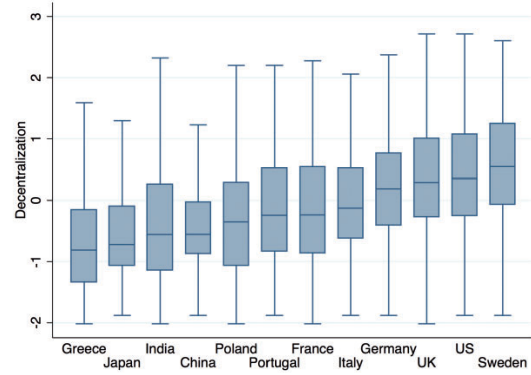


FIGURE III
Decentralization by Country (across Firms)

The graph shows levels of the z-scored decentralization index by country, measured as the average plant manager’s degree of autonomy over hiring, investment, products, and prices, at the 25th percentile (bottom line of the box), median (middle line of the box), and 75th percentile (top line of the box), as measured in the CEP organizational survey. Upper and lower adjacent values are also shown. The number of firms surveyed in each country is as follows: Greece = 183; Japan = 120; India = 397; China = 537; Poland = 222; Portugal = 145; France = 217; Italy = 96; Germany = 327; United Kingdom = 557; United States = 638; Sweden = 216.

(b) Figure III: decentralization by country, across firms

7.3 Figures IV and V: external validation of the decentralization measure

Read:

- Figure IV compares the paper’s firm-level decentralization measure with Hofstede’s Power Distance Index. Because the x-axis is scaled so that higher values mean lower power distance, the positive relationship validates the decentralization measure.
- Countries with lower hierarchical distance, such as Sweden, the United States, and the United Kingdom, have more decentralized firms.
- Figure V compares firm decentralization with Arzaghi and Henderson’s fiscal decentralization index.
- The positive relationship suggests that countries with decentralized firms also tend to have decentralized public-sector structures.

Economic message: The validation figures show that the survey measure is not merely an artifact of the interview process. It lines up with independent cross-country measures of hierarchy and decentralization.

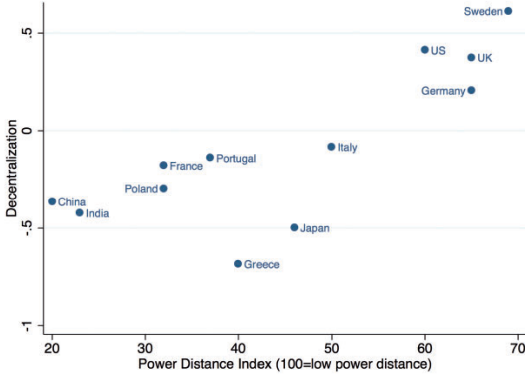


FIGURE IV
Decentralization and Power Distance Index by Country

The y-axis denotes levels of the z-scored decentralization index by country, measured as the average plant manager's degree of autonomy over hiring, investment, products, and prices by country, as measured in the CEP organizational survey. The x-axis is Hofstede's (2001) Power Distance Index downloaded from www.geerthofstede.nl/dimension-data-matrix. The number of firms used to build the decentralization measure in each country is as follows: Greece = 183; Japan = 120; India = 397; China = 537; Poland = 222; Portugal = 145; France = 217; Italy = 96; Germany = 327; United Kingdom = 557; United States = 638; Sweden = 216.

(a) Figure IV: decentralization and power distance

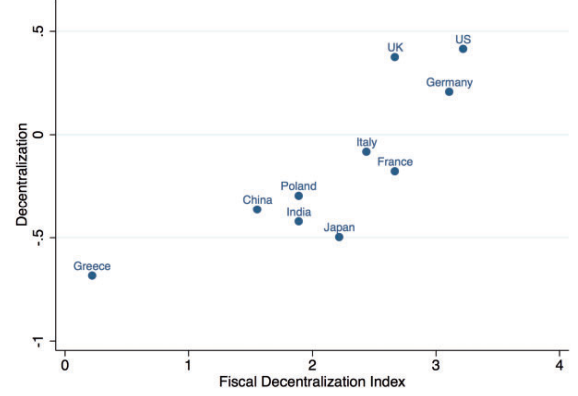


FIGURE V
Firm and Fiscal Decentralization by Country

The y-axis denotes levels of the z-scored decentralization index by country, measured as the average plant manager's degree of autonomy over hiring, investment, products, and prices by country as measured in the CEP organizational survey. The x-axis is Arzaghi and Henderson's (2005) Fiscal Decentralization Index. The number of firms used to build the decentralization measure in each country is as follows: Greece = 183; Japan = 120; India = 397; China = 537; Poland = 222; Portugal = 145; France = 217; Italy = 96; Germany = 327; United Kingdom = 557; United States = 638; Sweden = 216.

(b) Figure V: firm and fiscal decentralization

7.4 Tables I and II: baseline trust results and multinational identification

Read:

- Table I reports the baseline relationship between trust and decentralization.
- Column (1) shows a large unconditional trust coefficient of 1.231. Column (4), with country dummies, regional controls, industry dummies, and survey controls, still shows a positive and significant coefficient of 0.596.
- Plant skills, firm size, and plant employment share are also positively related to decentralization.
- Columns (5) and (6) split by whether the CEO is off-site or on-site. Trust is much stronger and significant when the CEO is off-site, consistent with the mechanism that trust matters more when monitoring is harder.
- Table II uses firms whose headquarters and plants are in different locations, especially foreign multinationals.
- Bilateral trust is positive and significant. In the demanding specification with country-of-origin and country-of-location dummies, the coefficient is 1.669.
- The IV coefficient using religious similarity as an instrument is 3.071, and the first-stage F-statistic is 28.56.

Economic message: Table I establishes the basic trust-decentralization relationship. Table II makes the evidence more credible by exploiting multinational structure, country-pair trust, and an instrument based on religious similarity.

Sample	(1) All	(2) All	(3) All	(4) All	(5) CEO off-site	(6) CEO on-site
Trust	1.231*** (0.440)	0.589*** (0.327)	0.669*** (0.365)	0.586*** (0.219)	0.558*** (0.380)	0.288*** (0.331)
Trust measured in CHQ region/country of location						
Rule of law (country of plant location) (-2.5=low, 2.5=high)		0.580*** (0.071)	0.550*** (0.065)	0.122*** (0.030)	0.162*** (0.038)	0.090*** (0.030)
Plant skills			0.086*** (0.030)	0.047*** (0.028)	0.053*** (0.038)	0.024*** (0.038)
% plant employees with a college degree			0.022*** (0.025)	0.025*** (0.028)	0.028*** (0.038)	0.028*** (0.038)
Firm size			0.134*** (0.054)	0.098*** (0.026)	0.125*** (0.043)	0.055*** (0.030)
Plant employment			0.213*** (0.064)	0.084*** (0.029)	0.158*** (0.038)	0.212*** (0.045)
Plant employees as a % of firm						
Foreign multinational						
Dummy=1 if firm belongs to a foreign multinational						
Domestic multinational						
Dummy=1 if firm belongs to a domestic multinational						
Observations	3,655	3,655	3,655	3,655	3,655	3,655
Country of CHQ location controls (2)	No	Yes	Yes	Yes	Yes	Yes
Country of plant location dummies (11)	No	No	No	Yes	Yes	Yes
Region of plant location controls (2)	No	No	No	Yes	Yes	Yes
Industry dummies (148)	No	No	No	Yes	Yes	Yes
Other controls (37)	No	No	No	Yes	Yes	Yes
Clustering	CHQ location	CHQ location	CHQ location	CHQ location	CHQ location	CHQ location
Number of clusters	146	146	146	146	127	135

Notes: The dependent variable in all columns is the decentralization α -score index, measured by plant manager's autonomy over hiring, investment, products, and marketing. Estimation is by OLS with robust standard errors in parentheses. Standard errors clustered by the firm's headquarters region of location (country of origin if the plant belongs to a foreign multinational). "Country of CHQ location controls" is the log of GDP per capita and population in the country of CHQ location. "Region of plant location" controls are the log of GDP per capita and population in the region where the plant is located. "Industry dummies" are two-digit SIC dummies. "Other controls" include a dummy for whether the firm is publicly listed, a dummy for whether the CEO is on the same site as the plant ("CEO on-site"), the fraction of managers native of the country of plant location, and "Noise controls" (these include 44 interviewee dummies, 6 dummies to control for the day of the week the interview took place, an interview reliability score, the manager's seniority and tenure, and the duration of the interview). Regressions weighted by the share of World Values Survey respondents in the region within the country. *, significant at 10%; **, significant at 5%; ***, significant at 1%.

(a) Table I: decentralization and trust

Sample	(1) CHQ in different region/country	(2) CHQ in different region/country	(3) Foreign multinationals	(4) Foreign multinationals (bilateral trust data available)	(5) Foreign multinationals (bilateral trust data available)	(6) Foreign multinationals (bilateral trust data available)	(7) Foreign multinationals (bilateral trust data available)	(8) Foreign multinationals (bilateral trust data available)
Estimation method	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS
Trust	0.609*** (0.270)	0.120*** (0.284)	0.198*** (0.310)	0.609*** (0.284)	-0.219*** (0.471)	1.785*** (0.819)	1.669*** (0.789)	3.071*** (1.255)
Trust measured in CHQ region/country								
Trust of people from country of origin for people in country of location								0.491*** (0.241)
Religious similarity								0.421*** (0.241)
Observations	1094	1094	881	422	422	422	422	422
Country of CHQ location controls (2)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country of plant location dummies (11)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region of plant location controls (2)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies (23)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other controls (37)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region of plant location dummies (11)	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country of CHQ location dummies (13)	No	No	No	No	No	Yes	Yes	Yes
Clustering	CHQ location	CHQ location	CHQ location	CHQ location	CHQ by plant location	CHQ by plant location	CHQ by plant location	CHQ by plant location
Number of clusters	92	92	33	16	97	97	97	97
First stage F-stat								Religious similarity 28.98

Notes: Dependent variable is the decentralization α -score index, measured by plant manager's autonomy over hiring, investment, products, and marketing. Columns (1) and (2) include all firms whose CHQ is located in a different region within the same country, or in a different country; columns (3)-(8) include only foreign multinationals. Instrument is "religious similarity" between each country pair. Standard errors in parentheses are clustered as noted. "CHQ by plant location" indicates clustering within each country origin by country of location only. "TRUST" measures the percentage of individuals who agreed with the statement "most people can be trusted" in the geography of firm's CHQ region or country of location. "BILATERAL TRUST" measures the percentage of individuals who agreed with the statement "most people can be trusted" in the geography of firm's CHQ region or country of location. "RELIGIOUS SIMILARITY" measures the percentage of people from country of origin who report to "trust a lot" people living in the country of firm's location. "Country of CHQ location controls" is the log of GDP per capita and population in the country of CHQ location. "Region of plant location" controls are the log of GDP per capita and population in the region where the plant is located. "Industry dummies" are two-digit SIC dummies. "Other controls" include a dummy for whether the firm is publicly listed, a dummy for whether the CEO is on the same site as the plant ("CEO on-site"), the fraction of managers native of the country of plant location, and "Noise controls" (these include 44 interviewee dummies, 6 dummies to control for the day of the week the interview took place, an interview reliability score, the manager's seniority and tenure and the duration of the interview). Regressions weighted by the share

(b) Table II: exploiting differences in CHQ location

7.5 Tables III and IV: robustness and firm size

Read:

- Table III adds possible omitted mechanisms one at a time and then all together: management quality, individual pay incentives, family ownership, hierarchical religion, and competition.
- The trust coefficient remains positive and significant in every specification, ranging roughly from 0.565 to 0.694, and equals 0.618 when all extra controls are included.
- Management quality, bonus pay, and competition are positively associated with decentralization. Hierarchical religion is negatively associated with decentralization.
- Table IV tests the firm-size prediction. Regional trust is positively related to average firm employment; with country dummies the coefficient is 2.270.
- In the multinational sample, bilateral trust predicts larger mean employment, larger total employment, and more subsidiaries across country-pair cells.

Economic message: Table III shows that the trust result is not just a proxy for management, incentives, ownership, religion, or competition. Table IV connects the organization result to firm scale, consistent with the theory that trust enables delegation and growth.

Additional controls	(1) None—baseline	(2) Management quality	(3) Individual pay incentives	(4) Family ownership	(5) Hierarchical religion	(6) Competition	(7) All
Trust	0.589*** (0.223)	0.565** (0.223)	0.608*** (0.217)	0.580*** (0.217)	0.694*** (0.182)	0.583*** (0.175)	0.618*** (0.159)
Trust in CHQ region/country of location							0.159*** (0.036)
Management							0.419*** (0.119)
Bonus		0.377** (0.171)					0.054 (0.045)
Family management				-0.091 (0.065)			-0.004** (0.002)
Hierarchical religion					-0.004*** (0.002)		0.150*** (0.043)
Competition						0.088** (0.027)	0.088** (0.027)
Observations	3,655	3,655	3,655	3,655	3,655	3,655	3,655
Country of CHQ location controls (2)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country of plant location dummies (11)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region of plant location controls (2)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies (148)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other controls (37)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	CHQ location	CHQ location	CHQ location	CHQ location	CHQ location	CHQ location	CHQ location
Number of clusters	146	146	146	146	146	146	146

Notes: Dependent variable is the decentralization α -score index, measured by plant manager's autonomy over hiring, investment, products, and marketing. Estimation is by OLS with robust standard errors in parentheses. Standard errors clustered by the firm's headquarters region of location (country of origin if the plant belongs to a foreign multinational). TRUST measures the percentage of individuals who agreed with the statement "most people can be trusted" in the firm's headquarters region of location (country of origin if the plant belongs to a foreign multinational). MANAGEMENT is the French-level Bonin and Van Rossum (2007) management score. BONUS is the percentage of managerial compensation tied to individual bonus, and firm performance. FAMILY MANAGEMENT is a dummy equal to 1 if the firm is owned and run by family members. HIERARCHICAL RELIGION is the percentage of people belonging to a hierarchical religion in the region of plant location as recorded by the World Values Survey (see text). COMPETITION is a variable measuring the number of the firm's direct competitors, as perceived by the plant manager. 1 = no competitors, 1 = between 1 and 5 competitors, 2 = more than 5 competitors. "Country of CHQ location controls" is the log of GDP per capita and population in the country of CHQ location. "Region of plant location" controls are the log of GDP per capita and population in the region where the plant is located. "Industry dummies" are three-digit SIC dummies. "Other controls" include a dummy for whether the firm is publicly listed, a dummy for whether the CEO is on the same site as the plant ("CEO on-site"), the fraction of managers native of the country of plant location, and "Noise controls" (these include 44 interviewee dummies, 6 dummies to control for the day of the week the interview took place, an interview reliability score, the manager's seniority and tenure, and the duration of the interview). Regressions weighted by the share of World Values Survey respondents in the region within the country. 146 clusters included in the regression. **, significant at 5%; ***, significant at 1%.

(a) Table III: robustness

Dependent variable	(1) Ln(employees) Mean	(2) Ln(employees) Mean	(3) Ln(employees) Mean	(4) Ln(employees) Total	(5) Ln(subsidiaries) Total
Measure	All firms	All firms	Foreign multinational	Foreign multinational	Foreign multinational
Sample	All firms	All firms	Foreign multinational	Foreign multinational	Foreign multinational
Trust (region)	2.270*** (0.727)	2.270*** (0.826)			
Trust measured in firm's region of location					
Bilateral trust			2.200*** (0.965)	5.578*** (1.477)	3.318*** (0.828)
Trust of people from country of origin for people in country of location					
Observations	110	110	292	292	292
Country location dummies (11)	No	Yes	Yes	Yes	Yes
Country of origin dummies	n/a	n/a	Yes	Yes	Yes
Clustering	Region	Region	CHQ by plant location	CHQ by plant location	CHQ by plant location
Number of clusters	110	110	292	292	292

Notes: Columns (1) and (2) are based on data reported by all domestic firms (i.e., excluding foreign multinationals) located in a specific region, aggregated at the regional level. The dependent variable in columns (1) and (2) is the log of the mean number of employees reported by all firms in the region. The set of regions included in the sample coincides with that included in the organizational survey sample, and standard errors are clustered at the regional level. Columns (3) to (5) are based on data reported by all multinational subsidiaries located in a specific country, aggregated by country of location-country of origin pair (i.e., the mean number of employees of Japanese multinational subsidiaries in France would be one observation, the mean number of employees of Japanese multinational subsidiaries in the United Kingdom would be another). The set of country of location-country of origin pairs coincides with that included in the Eurobarometer bilateral trust survey, and standard errors are clustered at the country of origin-country of location level. In column (3) the dependent variable is the log of the mean number of employees by country of location-country of origin pair, in column (4) the dependent variable is the log of the total number of employees by country of location-country of origin pair, and in column (5) it is the log of the total of subsidiaries in every country of location-country of origin pair. Online Appendix A1 provides details on the sources of information used to build the employment measures at the regional level and at the country of location-country of origin level. TRUST measures the percentage of individuals who agreed with the statement "most people can be trusted" in the geography of firm's CHQ region or country of location. BILATERAL TRUST measures the percentage of people from country of origin who report to "trust a lot" people living in the country of firm's location. Regressions weighted by the share of World Values Survey respondents in the region within the country in first two columns. **, significant at 5%; ***, significant at 1%.

(b) Table IV: firm size and trust

8 Short summary

- The paper studies whether trust affects firm organization and productivity.
- The central outcome is decentralization from headquarters to plant managers.
- The authors collect original survey data on almost 4,000 manufacturing firms across 12 countries.
- Trust is measured using the World Values Survey and, for multinationals, bilateral trust between country pairs.
- The theory predicts that trust raises decentralization because CEOs are more willing to let plant managers solve problems.
- The theory also predicts that trust increases firm size because delegation saves CEO time and expands the feasible span of control.
- Empirically, firms headquartered in high-trust regions are more decentralized.
- The result holds with extensive country, region, industry, management, incentive, ownership, and survey controls.
- Multinational subsidiaries are more decentralized when parent-country people trust host-country people more.
- IV evidence using religious similarity supports the causal interpretation.
- Trust is also associated with larger firms and larger multinational presence.
- The aggregate implication is that trust may raise productivity by allowing efficient firms to grow and resources to reallocate toward them.

9 Conclusion

9.1 Core conclusion

Bloom, Sadun, and Van Reenen show that social capital has an organizational channel. Higher trust makes firms more willing to decentralize decision making from headquarters to plant managers. This result appears in cross-regional data, in multinational subsidiaries, and in IV specifications using religious similarity as an instrument for bilateral trust.

9.2 Economic intuition

The economic intuition is that decentralization requires headquarters to rely on local managers. When trust is low, CEOs fear that delegated decisions will be wrong, opportunistic, or inconsistent with firm objectives. When trust is high, delegation is less costly. The CEO can then focus on exceptional problems and supervise more plants and workers. Trust therefore increases both decentralization and firm size.

9.3 Incremental contribution revisited

The paper makes three linked contributions. First, it creates a new cross-country data set on firm organization. Second, it provides a theory that connects trust to delegation and firm scale. Third, it uses multinational and bilateral-trust evidence to move beyond simple country-level correlations. This turns the broad claim that “culture matters” into a concrete firm-level mechanism.

9.4 Implications for productivity and development

The results help explain why low-trust economies may have smaller firms and weaker reallocation. If CEOs cannot delegate, productive firms face a managerial-capacity constraint and cannot expand as easily. This reduces the ability of efficient firms to absorb labor and capital from less efficient firms. In this sense, trust can shape aggregate productivity through the firm-size distribution.

9.5 Limitations

The evidence is not a randomized experiment. Trust may still proxy for broader cultural or institutional features, even though the paper controls for many of them. The decentralization measure is based on survey responses and contains measurement error. The IV strategy using religious similarity is persuasive but may capture broader cultural proximity, not trust alone. Finally, the sample is medium-sized manufacturing firms, so external validity to services, microenterprises, and very large global firms remains an open question.

9.6 Takeaway

The enduring takeaway is that trust changes how firms are run. It allows authority to move downward in the hierarchy, lets CEOs manage larger organizations, and may improve aggregate productivity by enabling better reallocation. The paper therefore provides a micro-organizational foundation for the macro relationship between social capital and economic performance.